

## Multiple Choice

1. **Answer:** C

*Options:* A) Females are three times more affected than males; B) It never happens in children or post menopausal women; C) Males from any socioeconomic background can be affected; D) Prevalence rates are 2% for female adolescents

*Note:* The correct answer highlights that Anorexia Nervosa can affect individuals regardless of gender or socioeconomic status, emphasizing the disorder's prevalence among males across diverse backgrounds.

2. **Answer:** D

*Options:* A) Type 2 diabetes; B) Cancer; C) Cardiovascular disease; D) All of the options given are correct

*Note:* Obesity is associated with an increased risk of developing various diseases, including diabetes, heart disease, and certain cancers, making all the options provided correct.

3. **Answer:** A

*Options:* A) May be helpful in diverticular disease; B) Are always easily fermented by colonic bacteria; C) Are usually derived from soluble fibre sources; D) Should be avoided in constipation

*Note:* Fibre supplements may be helpful in diverticular disease because they can increase stool bulk and promote regular bowel movements, which may reduce the risk of diverticulitis by decreasing pressure in the colon.

4. **Answer:** A

*Options:* A) Mucous cells secrete the glycoprotein mucin; B) Chief cells the hormone gastrin (which stimulates acid secretion).; C) Parietal cells secrete pepsinogen, the inactive precursor of pepsin; D) G Cells secrete hydrochloric acid

*Note:* Mucous cells in the stomach lining are specifically responsible for secreting mucin, a glycoprotein that forms mucus, which protects the stomach lining from acidic gastric juices and mechanical damage.

5. **Answer:** D

*Options:* A) Childhood stunting before 24 months of age affects short term cognitive performance; B) Childhood stunting before 24 months of age affects long term cognitive performance; C) Neither of the statements are correct; D) Both of the statements are correct

*Note:* The correct answer is valid because both statements provide accurate information that aligns with established principles in health sciences.

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* The statement is false because gut microbes convert L-carnitine into trimethylamine (TMA), which can then be converted into trimethylamine N-oxide (TMAO), a compound associated with an increased risk of cardiovascular disease, rather than choline providing protection against it.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* Parathyroid hormone increases plasma calcium concentration by promoting the absorption of calcium in the intestine, not decreasing it.

8. **Answer:** True

*Options:* A) True; B) False

*Note:* Leucine, methionine, and threonine are classified as essential amino acids because the human body cannot synthesize them, necessitating their intake through dietary sources.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* BMI only considers weight and height to calculate a numerical value, without accounting for the composition of that weight in terms of fat mass versus lean mass, leading to potential inaccuracies in assessing an individual's body fat levels.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* Irritable bowel syndrome (IBS) is a functional gastrointestinal disorder that can manifest in various forms, including constipation-predominant IBS, where individuals experience frequent constipation as a primary symptom.

## Long Form Response

11. **Answer:** The absorption of vitamins from the intestine involves various factors and mechanisms, including the type of vitamin, the presence of dietary fats, and the health of the digestive system. Fat-soluble vitamins (A, D, E, K) require dietary fats for effective absorption, while water-soluble vitamins (B and C) are absorbed directly into the bloodstream. Additionally, the intestinal lining and its ability to transport nutrients play a crucial role in this process. Understanding how vitamins are absorbed is essential for health because it helps individuals make informed dietary choices and address deficiencies that can lead to serious health issues. Overall, all of these factors contribute to the effective absorption of vitamins, highlighting the complexity of nutrition and its impact on well-being.

*Note:* correct because it identifies the critical factors influencing vitamin absorption,

such as solubility type and dietary fat presence, while emphasizing the importance of a healthy digestive system in facilitating this process for overall health.

12. **Answer:** Adipose tissue, commonly known as body fat, serves as the largest endogenous substrate source in the body, playing a crucial role in energy storage and metabolism. It stores excess energy in the form of triglycerides, which can be broken down and used for fuel when the body needs it, especially during periods of fasting or intense physical activity. Beyond energy storage, adipose tissue also helps regulate hormones that influence appetite and metabolism, thus impacting overall body weight and health. The balance of adipose tissue is significant; too little can lead to energy deficits, while excessive amounts can contribute to obesity and related health issues, such as diabetes and heart disease. Therefore, maintaining healthy adipose tissue levels is essential for overall health and well-being.

*Note:* correct because it accurately describes the dual role of adipose tissue in energy storage and hormonal regulation, emphasizing its significance as a major energy source and its impact on metabolism and overall health.